Chapter 4 – Activity Based Costing Part 2

Reasons for refining a cost system

Indirect costs a significant part of total cost and accurate allocation is necessary – “broad averaging” is insufficient

Increasing product diversity – each product having different activities resulting in consumption of different levels of indirect costs.

To make better decisions regarding pricing, promotion and cost.

To be more competitive

Activity based costing - rationale

Activity Based Costing (ABC) connects products to activities – that is, what products require what activities to be performed and how much.

Then it connects costs to activities – that is, what activities demand what resources and in what amounts (costs).

Using these two steps, it connects costs to products in a cause-effect manner

Activity based costing - method

The first step in ABC is to create cost pools for various activities.

Each cost pool is then categorized into a cost hierarchy

A cost hierarchy categorizes various activity cost pools on the basis of the different types of cost drivers, cost-allocation bases, or different degrees of difficulty in determining cause-and-effect relationships.

ABC systems commonly use a cost hierarchy with four levels to identify cost-allocation bases that are cost drivers of the activity cost pools.

Cost hierarchies

There are four levels of the cost hierarchy:	Output unit-level costs (related to the individual units of a product or service)
	Batch-level costs (related to a group of units)
	Product (or service)-sustaining costs (related to support a particular product or service without regard to the number of units or batches)
	Facility-sustaining costs (related to costs of activities that cannot be traced to individual products or services)

Activities

Individual activities are the fundamental source of indirect costs.

An activity is an event, task, or unit of work with a specified purpose—for example, designing products, setting up machines, operating machines, or distributing products. These are often referred to as cost drivers.

Activities and hierarchies

Some activities consume costs based on the number of units produced – indirect material in operating machines – unit level cost

Some activities consume costs based on the number of batches produced – machine set-up before batch production – batch level cost

Some activities consume costs regardless of the number of units or batches but per product – product design cost – product level cost

Other costs that are not affected by number of units or number of batches or number of products – factory rent for example – facility level cost.

Allocate

For each product identify what activities are required and allocate associated overheads based on the following method	Unit level activities – allocate costs based on unit level cost driver rates
	Batch level activities – allocate costs based on batch level cost driver rates
	Product level activities – allocate costs based on product level cost driver rates
	Facility level costs – This is the remaining cost – typically “broad averaged” based on any acceptable cost allocation base

Example

Touchsafe handles make three types of door handles – brass, chrome and stainless steel. Production takes 20, 15 and 10 hours to manufacture a batch of 1000 handles of each respectively. Additional data per 1000 as follows:

	Bronze	**Stainless steel**	**Brass**
Projected Sales units	30,000	50,000	40,000
Selling price	$40	$20	$30
Direct material	$8	$4	$8
Direct labor	$15	$3	$9
Manufacturing OH based on DLH Traditional costing system	$12	$3	$9

Example

Hours per batch of 1000 units

	Bronze	Stainless steel	Brass
Direct labor hours	40	10	30
Machine hours	25	25	10
Setup hours	1.0	0.5	1.0
Inspection hours	30	20	20

Total overhead costs and activity levels are as follows

	Overhead cost	Activity level
Direct labor hours		2,900 hours
Machine hours		2,400 hours
Setup hours and cost	$465,500	95 setup hours
Inspection hours and cost	$405,000	2,700 inspection hours
Total	$870,500	

Traditional costing

- Cost of Bronze handles per unit based on traditional costing

Cost	
Direct Material	$8
Direct Labor	$15
Manufacturing overhead based on direct labor hours (plantwide rate)	$12
Total	$35

- Profit margin of Bronze handles based on traditional costing: $40 – $35 = $5

ABC Costing Setup cost

ABC cost driver rates:

Projected units = 30,000

Number of batches based on 1000 per batch = 30

Setup cost per hour = $\frac{465{,}500}{95}$ = \$4,900

Each batch has 1 set up hour – total 30 setup hours for bronze

Setup cost for 30 setup hours = \$4,900 x 30 = \$147,000

Setup cost per unit = $\frac{\$147{,}000}{30{,}000}$ = \$4.90 per unit

ABC Costing Inspection cost

ABC cost driver rates:

Projected units = 30,000

Number of batches based on 1000 per batch = 30

Each batch has 30 inspection hours – total 30 x 30 = 900 inspection hours setup hours for bronze

Inspection cost per hour = $\frac{405,000}{2,700}$ = \$150

Inspection cost for 900 inspection hours = \$150 x 900 = \$135,000

Inspection cost per unit = $\frac{\$135,000}{30,000}$ = \$4.50 per unit

ABC Costing

- Cost of Bronze handles (per unit) based on ABC

Cost	
Direct Material	$8
Direct Labor	$15
Setup cost	$4.90
Inspection cost	$4.50
Total	

- Profit margin of Bronze handles based on ABC costing: $40 – $32.40 = $7.60

ABC versus traditional costing

ABC is generally perceived to produce superior costing figures due to the use of multiple drivers across multiple levels.

ABC is only as good as the drivers selected and their actual relationship to costs. Poorly chosen drivers will produce inaccurate costs, even with ABC.

Using ABC does not guarantee more accurate costs!

ABC is an alternate way to allocate costs. It is generally considered to be more accurate and more costly to implement.

Need for ABC - signals

Significant amounts of indirect costs are allocated using only one or two cost pools.

All or most indirect costs are identified as output unit-level costs.

Products make diverse demands on resources because of volume, process steps, batch size, or complexity.

Products that a company is well-suited to make show small profits whereas products that a company is less suited to make show large profits.

Operations staff has substantial disagreement with the reported costs of manufacturing and marketing products or services.

Other issues

Gain the support of top management and create a sense of urgency.

Create a guiding coalition of managers throughout the value chain for the ABC effort.

Educate and train employees in ABC as a basis for employee empowerment.

Seek small short-run success as proof that the ABC implementation is yielding results.

Recognize that ABC is not perfect (better costs but complex system).

Activity Based Management (ABM)

A method of management decision-making that uses ABC information to improve customer satisfaction and profitability.

We define ABM broadly to include decisions about pricing and product mix, cost reduction, process improvement and product and process design.